

Instructor Guide: PyCaret, ARIMA & SARIMA

PyCaret Overview

PyCaret is a low-code AutoML framework. It automates preprocessing, model comparison, tuning, evaluation, and deployment. Key functions include `setup()`, `compare_models()`, `create_model()`, `tune_model()`, `evaluate_model()`, `predict_model()`, `finalize_model()`, and `save_model()`.

PyCaret Workflow

Load dataset, run setup, compare models, create a selected model, tune it, evaluate it, finalize it, and save it.

Diabetes Dataset Example

Use the built-in diabetes dataset from PyCaret. Explain features, target variable, train/test split, and model comparison.

Time Series Fundamentals

Time series data consists of observations indexed by time. Components include trend, seasonality, cyclic behavior, and noise.

ARIMA Theory

ARIMA stands for AutoRegressive Integrated Moving Average. Parameters p , d , q represent autoregression, differencing, and moving average.

Stationarity

Stationarity means statistical properties remain constant over time. Differencing is often used to achieve stationarity.

ACF and PACF

ACF helps identify moving-average terms. PACF helps identify autoregressive terms.

ARIMA Practical Example

Train ARIMA on Airline Passengers data. Forecast future passenger counts and evaluate using MAE and RMSE.

SARIMA Theory

SARIMA extends ARIMA by incorporating seasonality through seasonal parameters P , D , Q , and s .

ARIMA vs SARIMA

ARIMA models trend. SARIMA models trend plus seasonality and is generally superior for monthly seasonal datasets.

Airline Passenger Dataset

Classic forecasting dataset with monthly passenger counts. Demonstrates strong seasonal patterns.

Evaluation Metrics

MAE, RMSE, and MAPE are common metrics used to evaluate forecasting performance.

Teaching Notes

Demonstrate visual plots, compare forecasts, discuss parameter selection, and explain practical business applications.

Lab Exercises

1. Diabetes Classification using PyCaret. 2. ARIMA forecasting. 3. SARIMA forecasting. 4. Compare forecast errors.

Interview & Viva Questions

Explain AutoML, compare ARIMA and SARIMA, define stationarity, describe ACF/PACF, and explain model tuning.